```
clc;
clear;
close all;

%% FIR FILTER PARAMETERS
b = [0.0625 0.2500 0.3750 0.2500 0.0625];
Fs = 1000;
N = 128;

x = [1 zeros(1,N-1)];
n = 0:N-1;

%% 1. DIRECT REALIZATION
yd = filter(b,1,x);

%% 2. CASCADE REALIZATION
% Factor the FIR polynomial into first-order sections
z = roots(b);
yc = zeros(size(x));
yc(1) = b(1);

for k = 1:length(z)
    yc = filter([1 -z(k)],1,yc);
end

yc = real(yc);

%% 3. SYMMETRY AND LINEAR PHASE
symErr = max(abs(b-fliplr(b)));
D = (length(b)-1)/2;

if symErr < 1e-12
    symmetry = 'Symmetric';
else
    symmetry = 'Not Symmetric';
end

%% 4. FREQUENCY, PHASE AND GROUP DELAY
[H,w] = freqz(b,1,1024);
phase = unwrap(angle(H));
gd = grpdelay(b,1,1024);

%% 5. NUMERICAL COMPARISON
err = max(abs(yd-yc));
dev = err/max(abs(yd))*100;

%% DISPLAY RESULTS
fprintf('\n====================================\n');
```

```
==================================
```

```
fprintf(' FIR FILTER REALIZATION RESULTS\n');
```

```
 FIR FILTER REALIZATION RESULTS
```

```
fprintf('====================================\n');
```

```
====================================
```

```
fprintf('Sampling Frequency = %d Hz\n',Fs);
```

```
Sampling Frequency = 1000 Hz
```

```
fprintf('Number of Samples = %d\n',N);
```

```
Number of Samples = 128
```

```
fprintf('Coefficient Symmetry = %s\n',symmetry);
```

```
Coefficient Symmetry = Symmetric
```

```
fprintf('Symmetry Error = %.3e\n',symErr);
```

```
Symmetry Error = 0.000e+00
```

```
fprintf('Group Delay = %.2f samples\n',D);
```

```
Group Delay = 2.00 samples
```

```
fprintf('Maximum Error = %.3e\n',err);
```

```
Maximum Error = 8.327e-16
```

```
fprintf('Percentage Deviation = %.3e %%\n',dev);
```

```
Percentage Deviation = 2.220e-13 %
```

```
%% FIRST 10 IMPULSE RESPONSE VALUES
fprintf('\n n       Direct       Cascade\n');
```

```
 n       Direct       Cascade
```

```
for k = 1:10
    fprintf('%2d     %10.6f   %10.6f\n', ...
        n(k),yd(k),yc(k));
end
```

```
 0       0.062500     0.062500
 1       0.250000     0.250000
 2       0.375000     0.375000
 3       0.250000     0.250000
 4       0.062500     0.062500
 5       0.000000     0.000000
 6       0.000000     0.000000
 7       0.000000     0.000000
```

```
 8       0.000000     0.000000
 9       0.000000     0.000000
```

```
%% CASCADE ZEROS
fprintf('\nCascade Realization Zeros:\n');
```

```
Cascade Realization Zeros:
```

```
disp(z);
```

```
  -1.0002 + 0.0000i
  -1.0000 + 0.0002i
  -1.0000 - 0.0002i
  -0.9998 + 0.0000i
```

```
%% OUTPUT PARAMETER TABLE
Parameter = {
    'Direct realization output'
    'Cascade realization output'
    'Impulse response'
    'Phase response'
    'Comparison of realizations'
};

Observation = {
    'Direct FIR filtering'
    'Cascade of first-order sections'
    'Equal to FIR coefficient sequence'
    'Linear phase'
    'Equivalent transfer-function behavior'
};

Value = {
    sprintf('[%.4f %.4f %.4f %.4f %.4f]',yd(1:5))
    sprintf('[%.4f %.4f %.4f %.4f %.4f]',yc(1:5))
    sprintf('[%.4f %.4f %.4f %.4f %.4f]',yd(1:5))
    sprintf('Group delay = %.2f samples',D)
    sprintf('Max error = %.3e',err)
};

Unit_Note = {
    'Amplitude'
    'Amplitude'
    'h[n]'
    'Samples'
    'Numerical precision'
};

Output_Table = table(Parameter,Observation,Value,Unit_Note);
```

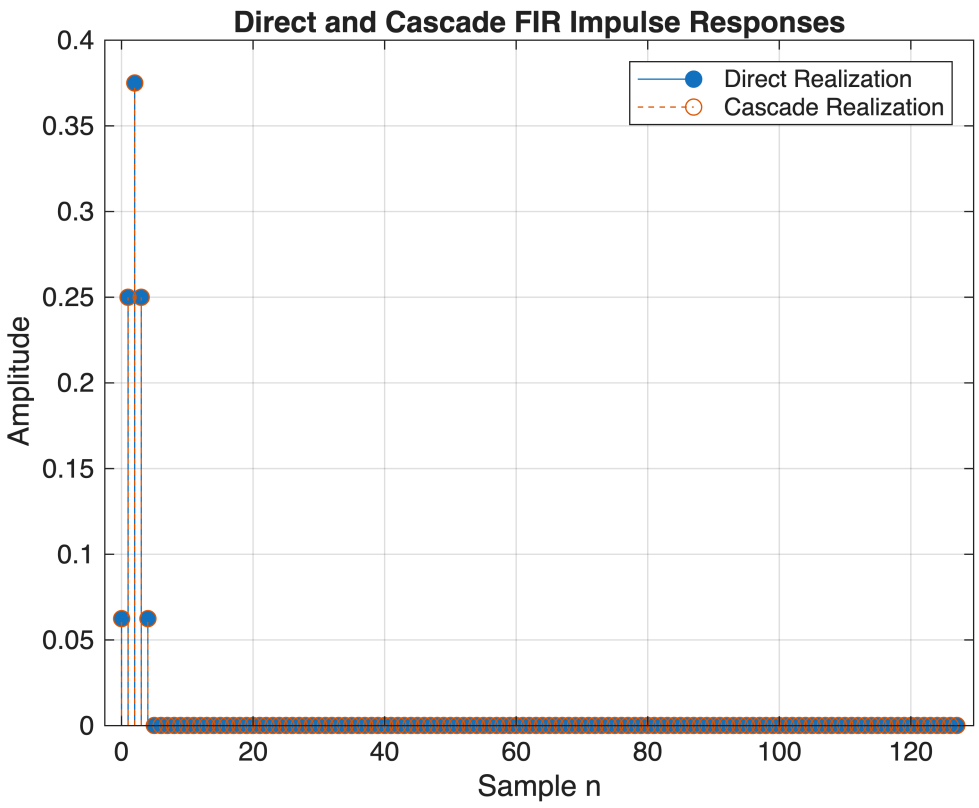
```
fprintf('\nOUTPUT PARAMETER TABLE\n');
```

```
OUTPUT PARAMETER TABLE
```

```
disp(Output_Table);
```

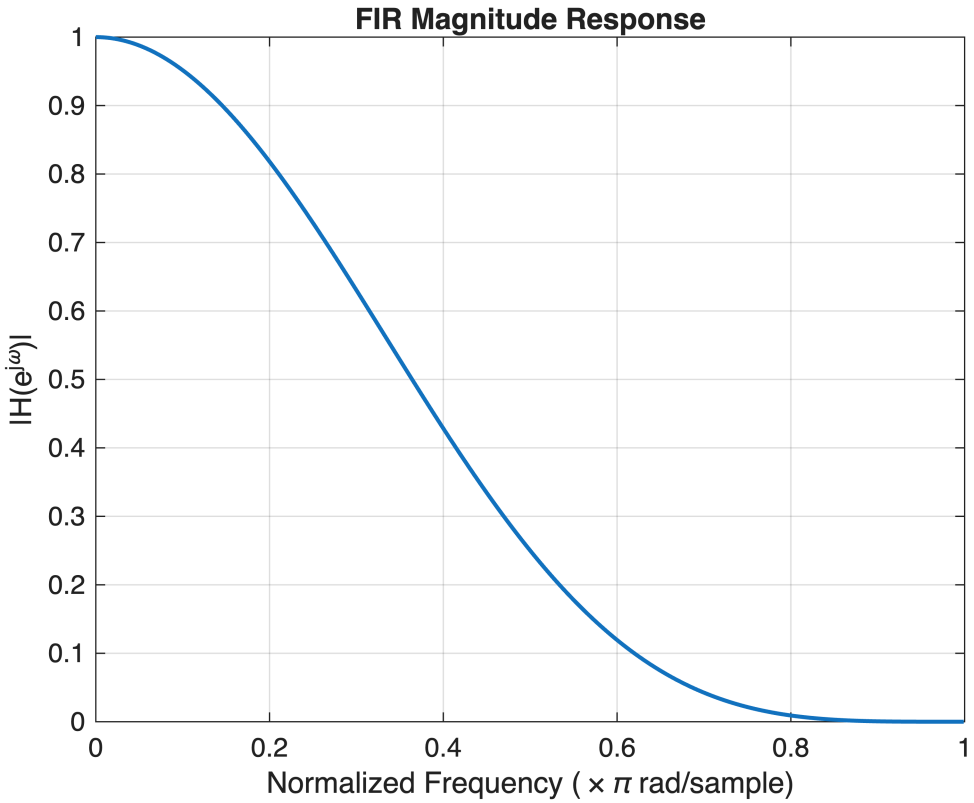
Parameter	Observation	Value
{'Direct realization output' }	{'Direct FIR filtering' }	{'[0.0625 0.2500 0.3750
{'Cascade realization output'}	{'Cascade of first-order sections' }	{'[0.0625 0.2500 0.3750
{'Impulse response' }	{'Equal to FIR coefficient sequence' }	{'[0.0625 0.2500 0.3750
{'Phase response' }	{'Linear phase' }	{'Group delay = 2.00 sa
{'Comparison of realizations'}	{'Equivalent transfer-function behavior'}	{'Max error = 8.327e-16

```
%% GRAPH 1: DIRECT VS CASCADE
figure;
stem(n,yd,'filled');
hold on;
stem(n,yc,'--');
grid on;
xlabel('Sample n');
ylabel('Amplitude');
title('Direct and Cascade FIR Impulse Responses');
legend('Direct Realization','Cascade Realization');
```

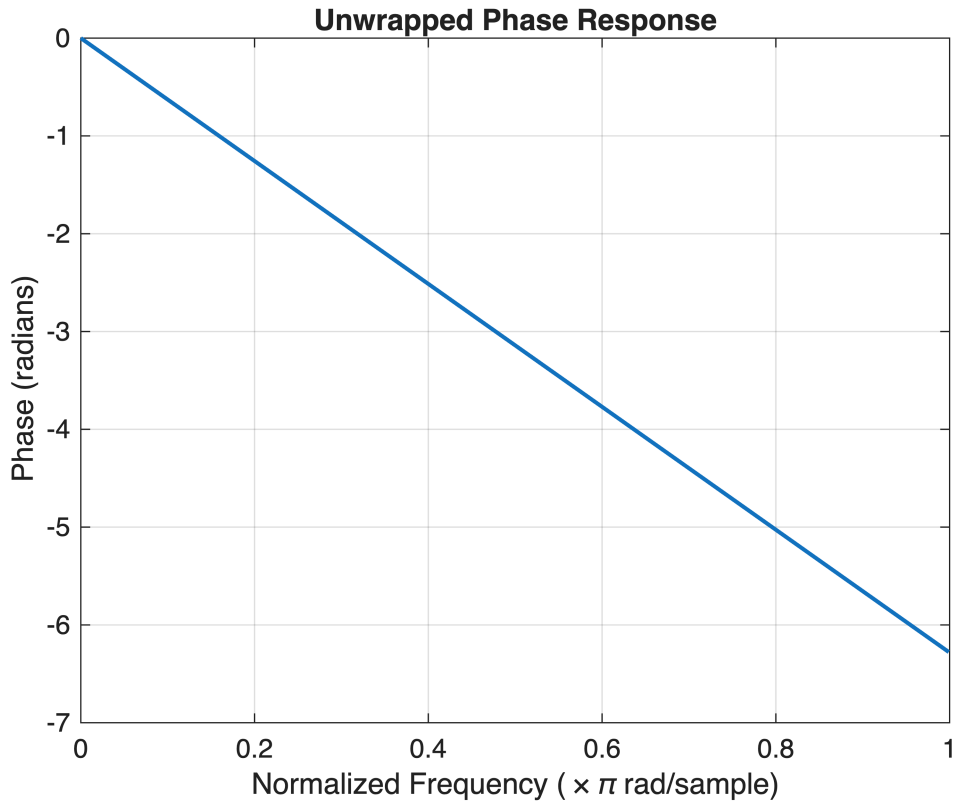



```
%% GRAPH 2: MAGNITUDE RESPONSE
```

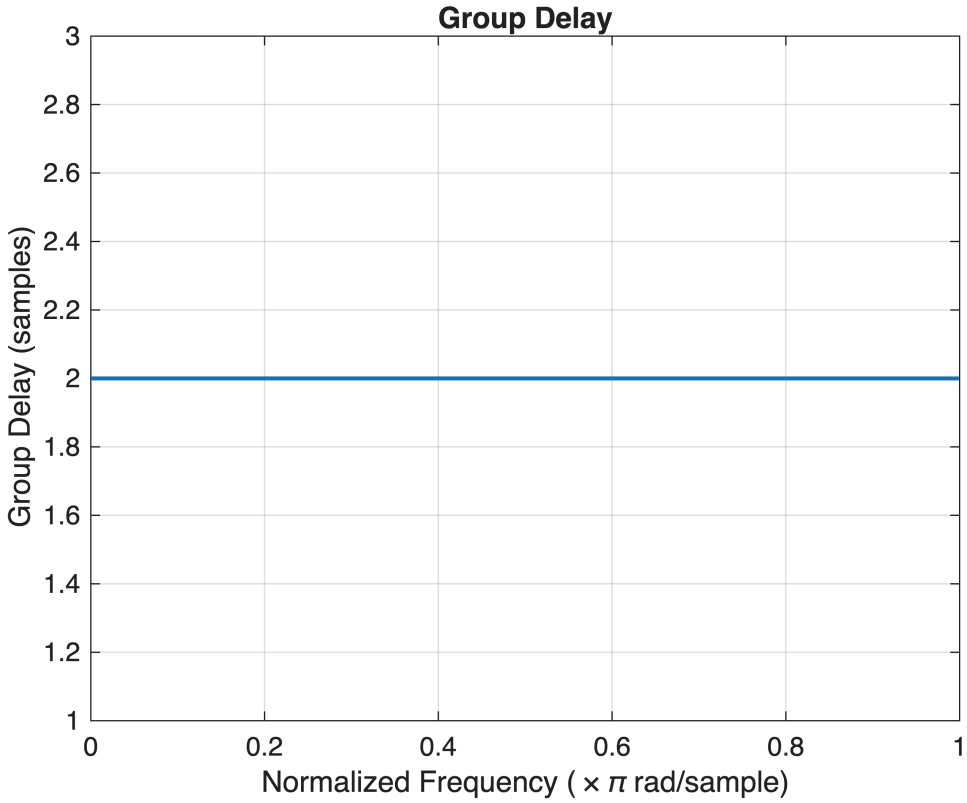
```
figure;
plot(w/pi,abs(H),'LineWidth',1.5);
grid on;
xlabel('Normalized Frequency (\times\pi rad/sample)');
ylabel('|H(e^{j\omega})|');
title('FIR Magnitude Response');
```



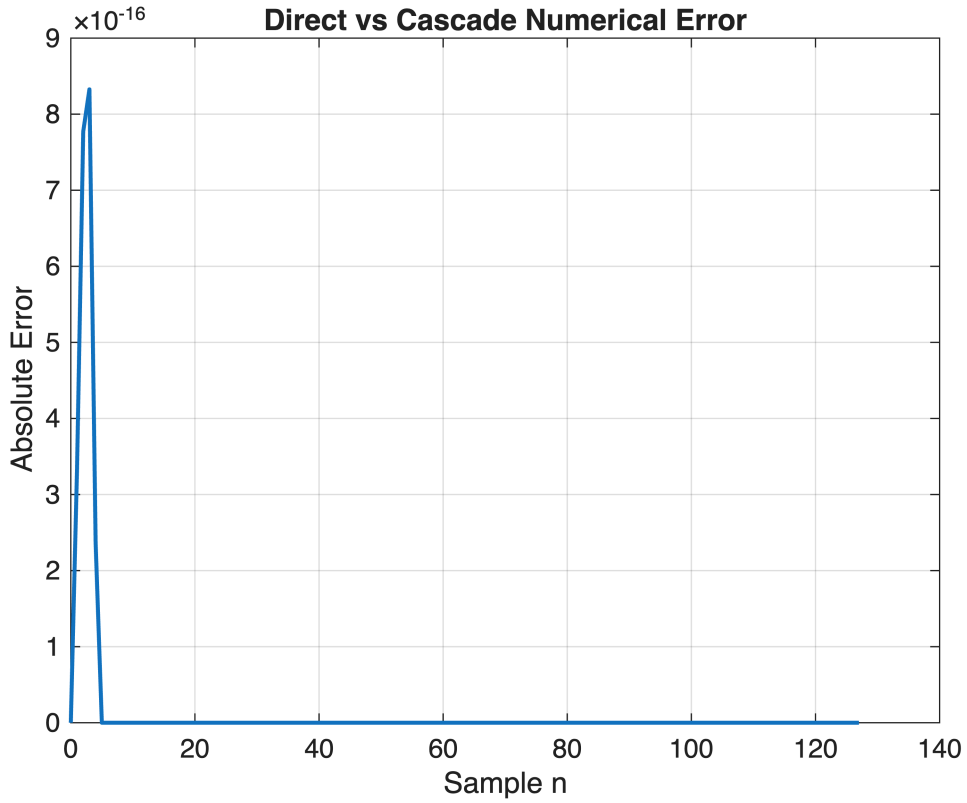

```
%% GRAPH 3: PHASE RESPONSE
figure;
plot(w/pi,phase,'LineWidth',1.5);
grid on;
xlabel('Normalized Frequency (\times\pi rad/sample)');
ylabel('Phase (radians)');
title('Unwrapped Phase Response');
```




```
%% GRAPH 4: GROUP DELAY
figure;
plot(w/pi,gd,'LineWidth',1.5);
grid on;
xlabel('Normalized Frequency (\times\pi rad/sample)');
ylabel('Group Delay (samples)');
title('Group Delay');
```




```
%% GRAPH 5: NUMERICAL ERROR
figure;
plot(n,abs(yd-yc),'LineWidth',1.5);
grid on;
xlabel('Sample n');
ylabel('Absolute Error');
title('Direct vs Cascade Numerical Error');
```




```
%% FINAL RESULT
fprintf('\n====================================\n');
```

```
==================================
```

```
fprintf('FINAL RESULT\n');
```

```
FINAL RESULT
```

```
fprintf('====================================\n');
```

```
==================================
```

```
fprintf('1. FIR filter realized using Direct structure.\n');
```

```
1. FIR filter realized using Direct structure.
```

```
fprintf('2. FIR filter realized using Cascade structure.\n');
```

```
2. FIR filter realized using Cascade structure.
```

```
fprintf('3. Coefficients are %s.\n',symmetry);
```

```
3. Coefficients are Symmetric.
```

```
fprintf('4. Linear-phase group delay = %.2f samples.\n',D);
```

```
4. Linear-phase group delay = 2.00 samples.
```

```
fprintf('5. Maximum realization error = %.3e.\n',err);
```

```
5. Maximum realization error = 8.327e-16.
```

```
fprintf('6. Percentage deviation = %.3e %%.\n',dev);
```

```
6. Percentage deviation = 2.220e-13 %.
```

```
fprintf('7. Direct and Cascade responses are equivalent.\n');
```

```
7. Direct and Cascade responses are equivalent.
```